WEBRUM-05: Error Analysis

> **Series:** WEBRUM — Web Real User Monitoring | **Notebook:** 5 of 10 | **Created:** March 2026 | **Last Updated:** 10/14/2026

Overview

JavaScript errors are among the most impactful issues affecting web application user experience. A single unhandled exception can break page functionality, prevent form submissions, or cause entire features to stop working. Dynatrace RUM captures JavaScript errors, XHR/fetch errors, and custom errors in real-time, enabling you to quantify error impact on real users.

Table of Contents

1. Error Types in RUM
2. Error Volume and Trends
3. Error Grouping and Top Errors
4. Error Impact Analysis
5. XHR and Fetch Errors
6. Rage Click Detection
7. Error-to-Session Correlation
8. Summary and Next Steps

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
RUM Enabled	Web applications with error capture active
Permissions	`storage:events:read`
Previous Notebook	WEBRUM-01: RUM Fundamentals

1. Error Types in RUM

Dynatrace captures several categories of browser-side errors:

Error Type	Description	Example
JavaScript error	Unhandled exception or runtime error	`TypeError: Cannot read property 'x' of undefined`
XHR error	Failed XHR/fetch request (HTTP 4xx/5xx or network error)	`HTTP 500 on POST /api/checkout`
Custom error	Errors reported via the RUM API (`dtrum.reportError()`)	`Payment validation failed`
CSP violation	Content Security Policy blocking a resource	`Refused to load script from 'http://evil.com'`
Resource error	Failed resource loading (images, scripts, stylesheets)	`Failed to load /assets/app.js`

Error Data Model

Error events are `user.events` records with `characteristics.has_error == true`:

Field	Description
`error.reason`	The error reason — RUM JavaScript reports a pre-defined set of values

Field	Description
`error.type`	Error classification: `exception`, `request` (failed request), `csp`, `reported` (custom reported error)
`error.source`	The error source, e.g. `fetch`, `console`
`page.detected_name`	Page on which the error occurred
`dt.rum.session.id`	Session containing the error
`start_time`	When the error occurred — `user.events` has no populated `timestamp`

The classic names `error.message`, `action.name` and `sessionId` read null on New RUM data.

> **Sources:** Errors — semantic dictionary (DT docs) — *"The error reason. RUM JavaScript reports a pre-defined set of values."*

```
// Error / navigation vocabulary corrected 08/12/2026 (New RUM):
//   filter type == "Error"  -&gt; filter characteristics.has_error == true
(11,909 events; identical
//                              population to isNotNull(error.type), whose
values are request/csp/exception)
//   error.message           -&gt; error.reason
//   user_action.type: "Load" -&gt; "hard_navigation", "RouteChange" -&gt;
"soft_navigation",
//                     "Xhr" -&gt; "same_view", "Custom" -&gt; "api"
(documented values, corrected 10/05/2026)
//   connection.type         -&gt; no web RUM equivalent
(network.connection.type is OneAgent for Mobile
//                              only; network.protocol.name is the OSI
protocol, e.g. "http")
// THE EVENT CHARACTERISTIC MATTERS AS MUCH AS THE FIELD: navigation-timing
fields
// (performance.dom_interactive, performance.load_event_end) live on
navigation events
// (characteristics.has_navigation) and are unpopulated on page summaries
// (characteristics.has_page_summary), so a page-summary filter silently
empties them.
// ttfb.* is the opposite — it lives on page summaries. Select with the stable
has_* flags, not
// characteristics.classifier (corrected 09/28/2026): the docs call classifier
"not intended for
// query usage" and Semantic Dictionary 1.349 removes it from the user-event
models.
// Session/performance field vocabulary corrected 08/12/2026 (New RUM).
Classic camelCase RUM
// names are null on New RUM data and fail silently. Verified against 3,261
user.sessions:
//   userType -&gt; dt.rum.user_type      userActionCount -&gt;
user_action_count
//   totalErrorCount -&gt; error.count    sessionId -&gt; dt.rum.session.id
//   hasSessionReplay -&gt; characteristics.has_replay
//   browserFamily -&gt; browser.name     osFamily -&gt; os.name
//   application -&gt; frontend.name (stable; an array on user.sessions, so
expand it before grouping)
//                  primary_tags.application exists only if your OpenPipeline
adds it (corrected 10/14/2026)
//   country/city/continent -&gt; geo.country.name / geo.city.name /
geo.continent.name
//   screen.width|height -&gt; browser.window.width|height
//   dom.interactive.time -&gt; performance.dom_interactive
//   load.event.time -&gt; performance.load_event_end
//   server.time -&gt; ttfb.value (page summaries;
web_vitals.time_to_first_byte where populated) —
```

```
//                  NOT ttfb.waiting_duration, which is the pre-request
wait/redirect phase
// TWO TENANT CAVEATS on the validation tenant, both of which leave a CORRECT
query empty:
//   * every session is dt.rum.user_type == "synthetic", so a real-user filter
matches nothing —
//     the documented values are "real_user" / "robot" / "synthetic"
(lowercase);
//   * geo.* is 0-populated, because synthetic traffic carries no geolocation.
// Field vocabulary corrected 08/12/2026 — this series targets **New RUM**,
but was written
// against names that are null on New RUM data, so these cells returned
nothing while erroring
// nowhere. Verified against 5,556,127 user.events records (schema 0.24.0,
javascript agent):
//   action.type == "Load"              -&gt; characteristics.has_navigation
== true
//                                        (was characteristics.classifier —
"not intended for query usage", SD 1.349)
//   action.type                        -&gt; user_action.type
(hard_navigation | soft_navigation | same_view | api)
//   action.name                        -&gt; page.detected_name
//   LCP / CLS                          -&gt;
web_vitals.largest_contentful_paint / web_vitals.cumulative_layout_shift
//                                        (stable — the fields Dynatrace's own
DQL examples use), falling back to
//                                        the experimental lcp.start_time /
cls.value (corrected 10/14/2026)
//   app.name                           -&gt; frontend.name (stable; an array
on user.sessions)
// UNITS DIFFER BETWEEN THE TWO FAMILIES. web_vitals.largest_contentful_paint
is a DURATION, so
// `/ 1ms` turns it into milliseconds; lcp.start_time is a PLAIN NUMBER
already in milliseconds,
// and dividing it by 1ms yields null. The CWV cells therefore read
// coalesce(web_vitals.largest_contentful_paint / 1ms, lcp.start_time) and
// coalesce(web_vitals.cumulative_layout_shift, cls.value), against the
2500/4000 ms and 0.1/0.25 thresholds.
// Why both: on the validation tenant every user event comes from the
synthetic engine, which fills
// lcp.* / cls.* but not web_vitals.* (0 of 243,494 page summaries in 30 days,
10/14/2026). That
// is a property of synthetic-monitor data, not a sign that web_vitals.* is
rarely populated.
// Recent RUM errors — explore the data structure
fetch user.events, from:-1h
| filter characteristics.has_error == true
| fieldsKeep start_time, error.reason, error.type, error.source,
```

```
page.detected_name, frontend.name, dt.rum.session.id
| sort start_time desc   // user.events has no populated timestamp; sorting on
it does not order rows
| limit 20
```

2. Error Volume and Trends

Start by understanding overall error volume and how it changes over time.

```
// Error count by type over last 24 hours
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize error_count = count(), by:{error.type}
| sort error_count desc
```

```
// Error trend over 24 hours — hourly bucketed by error type
fetch user.events, from:-24h
| filter characteristics.has_error == true
| makeTimeseries error_count = count(), interval:1h, by:{error.type}
```

```
// Error volume by application — which apps have the most errors?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize {error_count = count(),
    unique_errors = countDistinct(error.reason),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{frontend.name}
| sort error_count desc
```

3. Error Grouping and Top Errors

Error messages often contain variable data (stack traces, URLs, IDs). Grouping by error message helps identify the most frequent issues.

```
// Top 15 errors by frequency — the most common error messages
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize {error_count = count(),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{error.reason, error.type}
| sort error_count desc
| limit 15
```

```
// Errors by page — which pages generate the most errors?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter isNotNull(page.detected_name)
| summarize {error_count = count(),
    unique_errors = countDistinct(error.reason)},
    by:{page.detected_name}
| sort error_count desc
| limit 10
```

4. Error Impact Analysis

Not all errors are equal. An error affecting 1 session is different from one affecting 10,000. Impact analysis quantifies the blast radius of each error.

```
// Error impact — errors ranked by number of affected sessions
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize {total_occurrences = count(),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{error.reason}
| sort affected_sessions desc
| limit 10
```

```
// Error rate per application — percentage of sessions with errors
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| expand frontend.name   // array on sessions: a multi-frontend session counts
once per frontend
| summarize {total_sessions = count(),
    error_sessions = countIf(error.count &gt; 0)},
    by:{frontend.name}
| fieldsAdd error_rate_pct = round(toDouble(error_sessions) /
toDouble(total_sessions) * 100.0, decimals: 2)
| sort error_rate_pct desc
```

> **Tip:** Focus on errors with high session impact rather than high occurrence count. A single user hitting a retry loop can generate thousands of occurrences from one session.

5. XHR and Fetch Errors

XHR/fetch errors indicate backend API failures impacting the frontend. These are particularly critical in SPAs where the UI depends entirely on API responses.

```
// Unit trap (08/12/2026): New RUM timing fields such as lcp.start_time and
ttfb.value
// are PLAIN NUMBERS already in milliseconds, not durations. `field / 1ms`
yields null on them —
// silently, so a chart of nulls looks like "no data". Compare and aggregate
them directly.
// XHR errors by action — identify failing API calls
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter error.type == "request"
| summarize {error_count = count(),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{error.reason, page.detected_name}
| sort error_count desc
| limit 15
```

```
// XHR error trend — are backend errors increasing?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter error.type == "request"
| makeTimeseries error_count = count(), interval:1h
```

6. Rage Click Detection

A **rage click** occurs when a user rapidly clicks the same element multiple times — a strong signal of frustration. This usually indicates:

- A button that appears clickable but is not responding
- A form submission that is silently failing
- A UI element that is loading too slowly
- A dead link or broken navigation element

RUM Classic records rage clicks on the user session (USQL `numberOfRageClicks`). **New RUM documents no rage-click field.** The Semantic Dictionary has no field named for rage clicks or frustration, and `type` — which older versions of this notebook filtered as `type == "RageClick"` — is the *entity* type, empty on user events, so that filter returned nothing on every tenant. Until Dynatrace documents a New RUM rage-click signal, use Session Replay (WEBRUM-07) and the error-session correlation below to find frustrated users.

> **Dictionary:** no field matching `contains(name, "rage_click")`, `"rageclick"`, `"rage.click"` or `"frustrat"` in `dt.semantic_dictionary.fields`, read 10/05/2026 (control: the same query returned `network.connection.type` and `network.connection.subtype`); `type` (`stable`) — *"The entity type."*

7. Error-to-Session Correlation

Understanding how errors correlate with session outcomes (bounce, low engagement, failed conversion) helps prioritize fixes.

```
// Compare sessions with and without errors — engagement impact
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| fieldsAdd has_errors = if(error.count &gt; 0, "With Errors", else: "No
Errors")
| summarize {session_count = count(),
    avg_actions = avg(user_action_count),
    avg_duration_sec = avg(duration / 1s),
    bounce_rate = countIf(user_action_count == 1)},
    by:{has_errors}
| fieldsAdd bounce_pct = round(toDouble(bounce_rate) / toDouble(session_count)
* 100.0, decimals: 1)
```

```
// Errors by browser — are certain browsers more error-prone?
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| filter isNotNull(browser.name)
| summarize {total = count(),
    with_errors = countIf(error.count &gt; 0)},
    by:{browser.name}
| fieldsAdd error_rate = round(toDouble(with_errors) / toDouble(total) *
100.0, decimals: 1)
| filter total &gt; 10
| sort error_rate desc
| limit 10
```

8. Summary and Next Steps

In this notebook, we covered:

- **Error types** — JavaScript errors, XHR errors, custom errors, resource errors
- **Error volume and trends** — Tracking error rates over time
- **Error grouping** — Identifying the most frequent error messages
- **Impact analysis** — Measuring how many sessions each error affects
- **XHR/fetch errors** — Backend API failures visible from the frontend
- **Rage clicks** — a RUM Classic signal with no documented New RUM field; use Session Replay instead
- **Error-session correlation** — How errors impact engagement and bounce rates

Next Steps

- **WEBRUM-06: Performance Analysis** — Page load waterfall and performance bottleneck identification
- **WEBRUM-07: Session Replay** — Visually investigate error-impacted sessions

References

- Configure error detection for web applications (DT docs)
- User actions in RUM Classic (DT docs)
- User events — semantic dictionary (DT docs) — *"Used for internal optimization when storing the data and not intended for query usage."*
- Semantic Dictionary changelog 1.349 (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.